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Adaptive Interest Modeling Improves Content Services at the Network Edge

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***Abstract*—Content-based mobile ad-hoc networking (MANET) can be rapidly deployed in emergency and tactical situations at the network edge where fixed infrastructure is lacking. How to get the relevant content to the right user quickly, in the midst of network disruptions and resource constraints, is a key research challenge. Our insight is that if the network has some notion of users' interests and information needs, it can take proactive actions in order to make the network serve the users better. We have developed adaptive interest modeling (AIM) to capture and model user information needs at the network layer. AIM works by continuously monitoring network events at each node, and incrementally processing them to build an interest model (IM) for the node. One key contribution of our approach is to anchor AIM at the content-based network layer, which allows all upper level mobile applications to benefit without modification. This adaptive IM can enable many user-aware features including content prefetching and IM sharing. Our unique IM-enabled prefetching is based on recognizing user situations. IM sharing is a novel and efficient way of automatically keeping users up to date with each other in tactical scenarios. We implemented AIM as well as the IM-enabled prefetching and IM sharing features in a content-based MANET prototype. We evaluated these features in a network emulation environment. IM-enabled prefetching significantly reduces response time while increasing data availability at the same time. When combined with IM sharing, additional sizable reduction in response time is achieved.**

Keywords—user modeling; interest; MANET; network layer; content availability; prefetch; sharing

I. INTRODUCTION

As mobile computing technology rapidly advances and mobile devices become increasingly popular, more and more content is being generated by mobile devices such as mobile phones, tablets, and laptops. In addition, mobile devices can be rapidly deployed as a mobile ad-hoc network (MANET) at the network edge where fixed infrastructure is minimal or non-existent. With a large number of content-producing nodes, the edge network not only serves as communication medium, but also becomes a distributed data store with each node as a potential content producer and consumer [17]. This type of content-based edge networking plays an increasingly important role in both tactical and emergency response operations [6, 7].

Content-based edge networking requires efficient network services for disseminating, managing, and securing the distributed content. These content services can all benefit from the understanding and modeling of individual node's interests and information needs. For example, with content dissemination, the model of a node's interests can be utilized to refine any interest-based dissemination mechanism such as DIRECT (Disruption Resilient Content Transport) [15]. With content management, this understanding can make caching and prefetching more accurate and thus increase network performance while lowering the overhead [9, 17]. In the context of network security, models of node's interests can be employed to identify compromised nodes that behave anomalously [12].

With recent advances in content-based networking architectures, modeling of user interests at the network layer becomes feasible [13, 17]. These architectures provide a host-to-content abstraction in the form of publish/subscribe primitives. In particular, with the Haggle architecture, both the subscription interests and publication metadata are uniformly represented as attribute-value pairs [13]. As an extension to Haggle, the ICEMAN (Information Centric Mobile Ad-hoc Networking) architecture is designed to support distributed situational awareness applications in tactical scenarios [17]. In both Haggle and ICEMAN, subscriptions (i.e., queries) represent the subscriber's explicit information needs whereas publications reflect implicit interests of both the publisher and the reader.

There are several advantages for placing user interest modeling function at the network layer in the content-based MANET. Firstly, the function can be leveraged as a cross-layer service by other network components such as content dissemination and management. Secondly, any upper-level application can transparently benefit from user interest-derived capabilities such as prefetching. Thirdly, there is no need to modify the application code thus more applications can be deployed quickly. Lastly, there are potentially significant network savings because a user's interests and content needs are likely to overlap among different concurrent applications.

In this paper, we describe our work on Adaptive Interest Modeling (AIM), which explicitly and dynamically models mobile users' interests at the network level in an ICEMAN prototype. AIM monitors and processes network events to infer user interests and build an adaptive interest model (IM) for each user. The IM can in turn enable proactive capabilities that improve the performance of the content-based edge network-

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ing. One important capability is IM-enabled content prefetching. Prefetching is a common antidote to network disruptions prevalent in tactical MANETs. Our prefetching is unique in two aspects. First, it is positioned at the network layer instead of the application layer. Second, it is based on user's dynamic interests rather than simple heuristics (e.g., data size) or access patterns.

Another capability is IM sharing, the ability to send part or all of the IM to neighbors. In tactical situations, IM sharing has several benefits to the warfighters. Firstly, it is a more efficient way to keep neighbors up to date than sharing the data objects directly because IM is light-weight and contains distilled information. Secondly, IM sharing helps to build a common operational picture (COP) at the edge for better situational awareness (SA) because a warfighter gets to know what others know quickly. Lastly, it helps training and orientation for the warfighter because a newcomer can be brought up to speed automatically in terms of understanding the situation by seeing others' IMs.

We have implemented AIM in the ICEMAN prototype and evaluated the performance of IM-enabled prefetching and IM sharing using a network emulator. The evaluation studies show that AIM improves the response time significantly and has a positive impact on the content availability, albeit at the cost of using more network bandwidth. Our key contributions are: 1) the AIM adaptation algorithm for building IMs at the network layer; 2) an IM-enabled prefetching algorithm for combating network disruptions and improving system performance; and 3) an IM sharing algorithm for enhancing SA in tactical edge network scenarios.

II. AIM Approach

AIM takes advantage of several key characteristics of the ICEMAN architecture for its functionality. First of all, like its predecessor Haggle, ICEMAN is an event-based architecture, in which multiple managers cooperate in a layer-less fashion to provide content-based services. Within ICEMAN, AIM can continuously monitor the event queue and process relevant events. Secondly, in ICEMAN, a data object consists of metadata, represented as a set of attribute-value pairs, and a payload, which is represented by a file. AIM only processes the metadata of an object associated with an event under the assumption that the payload is typically protected and unavailable. Lastly, ICEMAN adopts a declarative naming approach where both publishers and subscribers identify content through weighted attribute-value pairs with a similarity threshold. This greatly simplifies the AIM's task of capturing a user's interests from the metadata because it obviates the need to call upon computationally expensive natural language processing.

Fig. 1 shows a high-level view of the AIM' approach under the ICEMAN architecture. User and network events are continuously monitored at the network level. The events include user queries, subscriptions, publications, and shared information. These events are then processed by the adaptation algorithm to infer user's interests and produce an IM. Once built, the IM can enable powerful applications including content prefetching and IM sharing. As demonstrated by our experiments below, content prefetching and IM sharing complement each other and together they bring faster query response, higher content availability, and better SA.

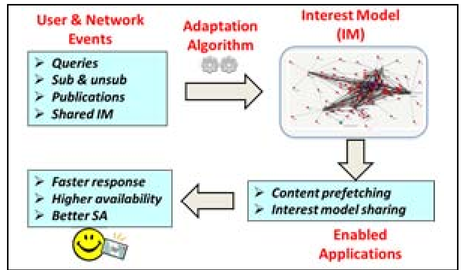

Fig. 1. The AIM approach.

A. *User Interests*

User interests can be grouped into three categories: explicit, correlated, and popular. Explicit interests are intentionally expressed by the user via a query or subscription. Correlated interests are inferred from content metadata if new interests co-occur with explicit interests in the same events. For example, a user may explicitly query for "IED", i.e., improvised explosive device. Later someone publishes a report tagged with both "IED" and "convoy". The term "convoy" becomes a correlated interest. Popular interests are also inferred from content metadata. They are frequently occurring information elements in content metadata associated with events in the neighborhood. Popular interests can reflect sudden situational changes on the ground in a tactical situation. For example, if in an ongoing operation, a large number of reports are tagged with the term "ambush", it is considered a popular interest.

Note that both correlated and popular interests are implicit interests because they are not overtly expressed by the user. Together with the explicit interests, they form a more complete picture of the user's information needs.

B. *User & Network Events*

As depicted in Fig. 1, at a high level the inputs for AIM are the user and network events generated in an ICEMAN implementation. In particular, AIM tracks and processes the following ICEMAN events to adapt the node's IM:

- Queries, subscriptions and un-subscriptions. Subscriptions express the user's explicit interests whereas un-subscription indicates the user is no longer interested in the associated contents. A query is a short-lived subscription event, which is terminated by an explicit un-subscription event.
- Publications. Contents published by the local node implicitly reflect the user's interests and information needs. Those published by other network nodes indicate interests on the network and are useful for deriving correlated and popular interests.
- Shared IM. This is an event generated by AIM's IM sharing function. A shared IM from a neighbor indicates that neighbor's situational awareness state and is useful for building a COP.

Table 1. AIM Adaptation Algorithm Description

1) *Create an empty interest model for the local node if it does not exist.*
2) *Extract the information elements from the metadata associated with the user event. For ICEMAN's attribute-value pairs, we convert each pair to a Term element*
3) *Age the current model by applying a decay function to all information elements in the interest model.*
4) *If an information element from the event already exists in the model, its weight in the model is positively or negatively reinforced. The amount of reinforcement is modulated by the source event type.*
5) *Otherwise, if the user event is positive, insert new information elements from the event into the interest model with a default weight modulated by their source event type.*

C. Adaptation Algorithm

The AIM adaptation algorithm is used for modeling the user's interests. It is based on the Reinforcement and Aging Modeling Algorithm (RAMA), which was developed in our previous research programs and was demonstrated to be effective in a formative evaluation study conducted by National Institute of Standards and Technology (NIST) [2, 11].

The AIM adaptation algorithm is described in Table 1. One key characteristic of the tactical edge is the pace at which the situation changes. Accordingly, user interests and information needs also change rapidly. To capture the changing user interests, the adaptation algorithm continually updates the IM with incoming user and network events by applying a reinforcement mechanism and a decay function. The reinforcement mechanism processes user events differently based on their polarity, i.e., positively or negatively reflecting user's interests. With positive events (e.g. subscription or publication) that express user's interests, the reinforcement mechanism will increase the importance or weight of the information elements contained in them. On the other hand, with negative events (e.g., unsubscription) that indicate user's lack of interest, the weight of the contained information elements will decrease. The decay function operates concurrently with the reinforcement mechanism. It causes the weights of information elements to gradually decrease over time. One advantage of the adaptation algorithm over many other machine learning methods is the ability to process the events incrementally. As a result, our algorithm can model users continuously and in real time.

The time complexity of RAMA is linear to the number of relevant network events in the worst case. RAMA processes events incrementally as they arrive. Computing both reinforcement and decay for an event take linear time to the size of the IM because the number of potential interests in the metadata of a single event is small and bounded by a constant. The size of the IM grows at worst case linearly with number of the events. The worst case occurs when the all potential interests in all events are unique. If we keep the IM size constant by removing interests with low important weights, RAMA can achieve constant time. Similarly, the space complexity has worst case linear to the size of incoming events and can be made to constant by keeping the size of the IM constant by non-important interests.

Table 2. An Example of Interest Model

{InterestModel:identifier=me,
numReportedEvents=2,
facet1={facet; ID=1; retired=0; Terms=
[(Term: IED, 0.82)]
[(Term: convoy, 0.89)]
; Elements=
[(OntologyNode: <urn:swat:reports-20111031.owl#IEDAlertReport>, 0.65)]
[(OntologyNode: <urn:swat:reports-20111031.owl#PositionReport>, 0.55)]}

Table 3. Example Profile Based on the Battle Drill for IED

{InterestModel:identifier=IED Profile,numReportedEvents=1,
facet1={facet; ID=1; retired=0;
Terms=
[(armoredvehicle,1)]
[(convoy, 1)]
[(explosive, 1)]
[(explosion, 1)]
[(wires, 1)]
[(ordinance, 1)]
[(smoke, 1)]
[(light, 1)]
[(wounded, 1)]
[(bomb, 1)]}

D. IM Representation

In the AIM context, the IM represents a dynamic and cohesive set of information elements that a user may be interested in. It is automatically created and continually adapted by the RAMA adaptation algorithm. An IM is comprised of one or more facets, which describe different aspects or time-sensitive phases of a user's interests. In tactical operations, the notion of time-based facets is well suited for capturing the swift changes in situation and information needs. A facet consists of a set of information elements, each of which represents some dimension of a user's interests. Each element carries a weight to indicate its degree of importance. The information element may take different forms including terms, named entities, ontological entities, topics and relationships. In particular, a term represents a user interest dimension corresponding to a word or phrase. With the ICEMAN architecture, terms may be readily extracted from attribute-value pairs in the content metadata associated with user and network events.

Table 2 shows an example of an IM generated by the AIM adaptation algorithm applied to a couple of subscription and publication events. In this example, the IM contains one facet with two term elements ("IED" with weight 0.82 and "convoy" with weight 0.89). It also contains two ontological elements ("IED Alert Report" with weight 0.65 and "Position Report" with weight 0.55).

The space of potential interests from the metadata overtime can be large. In order to scale, the weight associated with an interest can be used to maintain the IM at reasonable size by removing non-important interests from the model.

We use the cosine similarity metric based on vector space model to compare two IMs [14]. Our implementation of cosine similarity takes quadratic time on the size of the IM with linear space complexity.

E. Content Prefetching

IM offers different options for content prefetching. One option is to use the information elements in the IM directly. This is done by first automatically generating a query based on the top-weighted elements and then using it for fetch data on the user's behalf [3]. Another interesting and unique type of prefetching can be performed if we have domain knowledge about the user's roles or tactical operations. It is called the IM-enabled Profile-based Prefetching (IPP), where a profile refers to a common set of information needs and requirements for a given role or a specific operational situation. A battle drill-derived information profile (BDIP) is a profile built with a battle drill, which is "a collective action, executed by a platoon or smaller element, without the application of a deliberate decision-making process" according to the US Army Combined Arms Center and Fort Leavenworth Public Home Page (usacac.army.mil).

Table 3 is an example of a BDIP based on the battle drill related to IED. Note that the profile uses the same data structure as IM. It contains one facet, which in turn consists of ten terms that capture information needs related to the IED battle drill.

The IPP concept is illustrated in Fig. 2. We first create BDIPs (the blue circles in the figure) for common military operations such as IED, cordon, and search. AIM then builds and continuously updates the IM (the red dots in the figure) based on the user and network events. Whenever the current IM is updated, it is compared with the BDIPs to find the best match. We use the cosine similarity metric to compare an IM with a BDIP [14]. As the IM continues to evolve over time, a matched profile may become dissimilar and a new profile may become more similar. This can happen when the situation on the ground changes. For example, cordon and search operations may be interrupted by an IED explosion. In this case, the IM may initially match cordon and search profiles, but later switch to an IED profile. When a match is found, the information elements in the profile that are not covered by the IM will be used for prefetching. Contents matching these elements are automatically fetched and cached when they become available and typically before they are requested by the user.

F. IM Sharing

IM sharing is a mechanism for a node to automatically share its IM with other nodes on the network. There are two triggers for IM sharing: a) upon radio contact with a neighbor; and b) when the IM changes significantly. Once triggered, IM sharing is accomplished by sending a new data object encoded with IM elements to connected neighbors. Upon receiving a neighbor's IM, the encoded information elements will be incorporated into the receiver's IM. A couple of mechanisms are devised to control the behavior of IM sharing. One is to control the scope of sharing. It makes sense to allow users belonging to the same organizational unit to share. To do this, every node is configured to belong to a particular organizational unit, e.g., a combat squad. Another mechanism is put in place to control how much of the IM to share with others. AIM can be configured to share the entire IM or a preset number of top weighted elements of the IM.

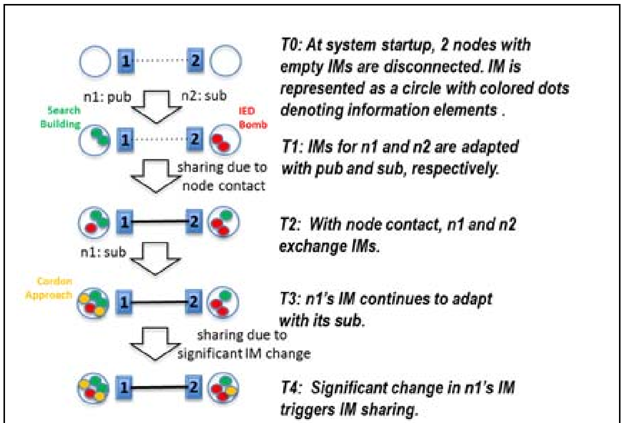

Fig. 3. Scenario: IM sharing improves SA

One benefit of IM sharing is that it can improve the network performance by working with prefetching. In particular, when IM is updated due to receiving shared IMs from neighbors, IPP may be automatically triggered to perform prefetching. Another benefit of IM sharing is that it can enhance SA by building a COP at the edge in a tactical situation. To illustrate the latter benefit, we describe a notional scenario (Fig. 3). Two nodes (n1 and n2) build IMs based on publication and subscription events. After a subscription event, for example, n2's IM has two terms, IED and Bomb in red. The two nodes share their IMs when they are in radio contact and when there's significant change in the IM. As a result of IM sharing, both nodes end up with similar colored dots in their IMs. In other words, IM sharing helps dynamically build a shared vision of what is happening in the neighborhood.

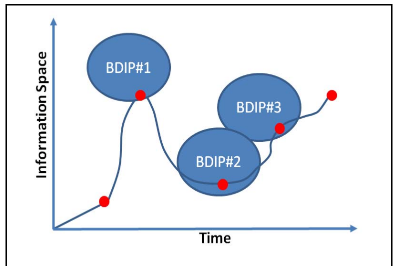

Fig. 2. IM-enabled and profile-based prefetching.

III. AIM IMPLEMENTATION

AIM is implemented on top of ICEMAN with a kernel consisting of a set of managers interacting with a single event queue in a coordinated fashion [17]. The main functions of AIM are implemented as an ICEMAN manager called **the interest manager** written in C++. The interest manager includes three functionalities: 1) IM creation and maintenance; 2) IPP; and 3) IM sharing.

For IM creation and maintenance, the interest manger listens for selected network and user events such as subscription, un-subscription, and publication. These events are passed on to the **adaptation** class which implements the AIM adaptation algorithm that learns the IM. The **InterestModel** class represents the learned interests for a node. It is comprised of several **facets**, each describing one aspect or phase of a user's interests. That facet is represented by a set of weighted **elements**, each of which represents one specific interest dimension.

IPP is another functionality built into the interest manager. It is implemented by creating a virtual application node with the interests the interest manager wants to prefetch based on the profile that matches the current IM. The interest manger is responsible for management, parsing, and configuration of profiles used for IPP. For our testing and evaluation, we have manually created three profiles based on battle drills for IED, search, and cordon, respectively (Table 3). A profile is activated if its cosine similarity with the current IM exceeds a threshold defined as a parameter in the configuration file. When a profile is activated, the interest manager selects configurable number of top weighted elements from the profile to include in the node description, which is a special data object that carries the interests of a node and a summary of data objects stored in the node, and is exchanged with neighbors upon contact [13, 17].

IM sharing is implemented by sending a new data object with IM elements to connected neighbors. When the amount of IM change is greater than the configurable threshold, the IM will be shared. The IM change is defined as 1 minus the cosine similarity between the current IM and the state of the IM when it was last shared. The amount of IM sharing is configured by the parameter "limit", which indicates the max number of top-weighted elements of the IM to be shared. The scope of IM sharing is controlled by the configuration parameter "units", a string representing the name of the organizational unit the node belongs to. When IM sharing occurs, the shared data object will carry the unit information as an attribute in the metadata. Upon receiving an IM from a neighbor, the local node's unit information will be compared to that carried on the incoming IM data object. Only if it matches, will the incoming IM be processed by the interest manager and incorporated into receiving node's IM. The incoming IM becomes input to the receiver's adaptation algorithm to update the current facet of the local IM.

IV. Evaluation

We have conducted two studies to evaluate the performance of AIM. In particular, we have assessed the performance impact of IM-enabled prefetching and IM sharing on key metrics for the ICEMAN prototype system.

A. Performance Metrics

The key network performance metrics that impact the warfighter at the tactical edge include **content availability**, **response time**, and **bandwidth consumption**. Content availability is measured by whether or not a relevant data object gets delivered to an interested node. Response time is measured by how long it takes the relevant information to reach the querying/subscribing node given that the relevant information has been published in the network. In addition, the cost of AIM is measured by the increase in network bandwidth consumption.

B. Test Description

The tests were run in the real-time CORE (Common Open Resource Emulator) 4.3 emulation environment [1]. Twelve warfighters (i.e., nodes) are on a mission operating in an area of 800 meters X 1100 meters. The radio range is set to 250 meters so that some of the warfighters are not within radio range of others. However, even if nodes are not in direct radio contact, nodes can share content and interests on the network via intermediate nodes (i.e., via multi-hop routing). The nodes move according to the random waypoint model, where the destination, speed and direction of any node are all chosen randomly and independently of other nodes [18]. The size of published contents ranges from 200K to 2MB. The data rate is set at 10Mbps. There are 6 publishers and 5 subscribers.

To understand the impact of the IM-enabled prefetching and IM sharing on content availability, response time, and bandwidth consumption, the performances of AIM turned off (AO), prefetching alone (PR), and sharing plus prefetching (SP) were compared under the following two network configurations:

- Forwarding turned off (NF) so that no multi-hop network connectivity exists.
- DIRECT turned on (DI) so that multi-hop routing is available.

By combining the two network configurations with AIM turned off, we have two baseline conditions: NF:AO and DI:AO.

Two experiments (Study 1 and Study 2) using slightly different tactical scenarios were conducted. In the Study 1 scenario, nodes publish interests by submitting queries and subscriptions according to Zipf-distribution, where the frequency of an interest is inversely proportional to its rank among all interests. Zipf-distribution has been shown to model the Internet content popularity well [8]. In the Study 2 scenario, some nodes publish interests late in the scenario. This type of behavior typically occurs when a network is disrupted and the warfighters do not have a COP or when new members join the team in the middle of an ongoing mission. The results from the two studies show similar overall trend. For brevity, we shall focus on the results from Study 2 and point out the performance differences between the two studies.

C. Study 2 Results

Response Time (Fig. 4 top panel)**.** With the NF:AO baseline, there is a significant (25%) improvement in response time for prefetching and a further 25% improvement for prefetching plus sharing (left three bars in the panel). Similarly with the DI:AO baseline, prefetching provides a very significant improvement in response times (more than 50%) and an additional 33% improvement is achieved with sharing added (right three bars in the panel). In Study 1, prefetching similarly reduced response time. However, adding IM sharing did not provide further benefit. The reason for this discrepancy on sharing between the two studies is caused by the difference in the scenarios. In Study 2, some nodes publish their interests late, and consequently can benefit from the prefetching resulted from previously built and shared IMs.

Content Availability (Fig. 4 middle panel). With the NF:AO baseline, prefetching provides a small improvement; but when AIM sharing is added, the improvement reaches a factor of nearly three. With the DI:AO baseline, AIM provides only a small improvement in content availability. The results on content availability is quite similar to the results achieved in Study 1.

Bandwidth Consumption (Fig. 4 bottom panel). When compared with the NF:AO baseline, prefetching alone caused very little increase in bandwidth consumption. However, when both prefetching and sharing are used, the bandwidth consumption increases significantly. With the DI:AO baseline, bandwidth consumption increased under prefetching and went up further when sharing was added. These results are nearly identical to those observed in Study 1.

D. Summary

The performance studies on IM-enabled prefetching and IM sharing capabilities indicate that the main benefits of AIM are a significant improvement in query response time (over 50%) and a modest increase in content availability. Naturally there is a trade-off between these benefits and the associated cost in terms of increased bandwidth consumption. Prefetching results in a relatively small increase in bandwidth consumption while providing large improvement in response times for both tests. IM sharing is more bandwidth intensive and provides no response time improvement in Study 1. However, in Study 2, which is more indicative of a disruptive environment, IM sharing provides significant incremental response time improvement over and above prefetching.

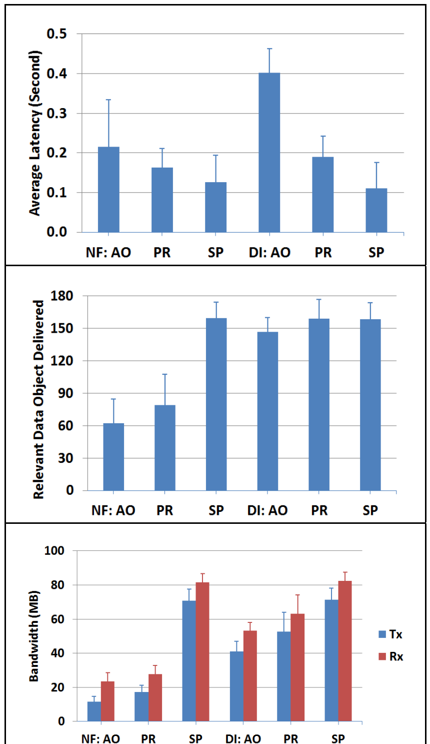

Fig. 4. Study 2 results. Top: AIM reduces query response time; Middle: AIM increases content availability; Bottom: AIM consumes more bandwidth. Legend: NF=No Forwarding, DI=DIRECT, AO=AIM Off, PR=Prefetching, SP= Sharing + Prefetching.

V. RELATED WORK

In the mobile computing domain, user interests are typically tracked by applications. Explicit modeling of user interests at the network level has not been actively researched due to the lack of application knowledge [8, 10, 16]. With recent content-based networking architectures such as Haggle and ICEMAN, we were able to implement AIM at the network layer to enable proactive content services [13, 17].

In disruptive networking environments, prefetching has commonly used for reducing latency. Effective prefetching requires good understanding of user interests to select the right data objects to fetch proactively. Some applications use simple but suboptimal heuristics (e.g., prefetching everything or those with size less than 32KB) to determine what objects to prefetch [8]. Other applications use mobility or data access patterns to predict what the user would likely to want next [5, 8, 10]. Our prefetching approach should be more precise and powerful because the IM is based on the semantics of user's behaviors.

Prefetching is often implemented by mobile applications such as a wireless web browser client [16]. Interestingly, Higgins et al. developed a system level interface to support prefetching for mobile applications (e.g., email or news reader) [8]. The interface depends on the application to provide hints of possible future accesses. We implemented the IM-enabled prefetching at the network layer with no dependency on the mobile applications that enjoy its benefit.

VI. CONCLUSION

We have developed the AIM functionality for content-based mobile edge networking. It is implemented as a software component that runs at the network layer in the ICEMAN prototype. AIM applies a light-weight machine learning algorithm to network queries, subscriptions, and publications to create and maintain an IM for each node. The IM can be used as the basis for prefetching information that "matches" the IM in anticipation of the user requesting that information. The IM can also be shared among warfighters to improve SA in tactical scenarios. Our experimental results show that AIM can deliver very significant reductions in query latency and measureable improvements in content availability in a disruptive tactical edge network.

The evaluation scenarios provided useful performance measures and were tactically relevant, but further experiments are needed to study the impact of large number of nodes and more realistic mobility models. Other applications of AIM would also be worthwhile to study in the future, such as the use of IM to influence ICEMAN's caching strategies. There is also potential to make our approach more adaptive and focused so that the bandwidth overhead is reduced, e.g., avoid prefetching/IM sharing when the network is already over-utilized. Algorithms exist for detecting network bursts and congestions. For example, average queue length, incoming data rate, and MAC contention can be combined for such detections.

For various reasons it can be useful to limit the construction of the IM to certain nodes, e.g. trusted nodes, potential endpoints, or nodes with more resources, but the performance implications need further investigation. Also the use of obfuscated or otherwise protected metadata will be an interesting avenue for future research.

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